

## NEW CENTURY, KS, USA

WMO: 724475

Lat: 38.832N

Lon: 94.890W

Elev: 331

StdP: 97.41

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 93909

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.5      | -12.7 | -20.2                         | 0.6 | -14.2 | -17.9 | 0.8 | -11.1 | 12.4                  | 3.9  | 11.5 | 2.7  | 4.0                   | 330  | 0.550 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 36.2            | 24.3 | 34.1 | 24.2 | 32.6 | 23.9 | 26.1                | 32.6 | 25.4 | 31.8 | 24.8 | 30.8 | 5.3                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.1                            | 19.8 | 29.7 | 23.6 | 19.1 | 29.1 | 22.8 | 18.3 | 28.2 | 82.8          | 32.8 | 79.8 | 31.6 | 76.7 | 30.8 | 29.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |      | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 11.3              | 10.1 | 8.9  | DB                         | -19.7 | 38.6 | 2.7                | 1.9                                                | -21.7     | 39.9 | -23.3      | 41.0 | -24.8      | 42.1 | -26.8      | 43.4 |  |
|                   |      |      | WB                         | -20.1 | 27.3 | 2.6                | 0.8                                                | -22.0     | 27.9 | -23.5      | 28.4 | -25.0      | 28.9 | -26.9      | 29.5 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.1   | -0.8  | 1.2   | 7.1   | 13.0  | 18.5  | 23.6  | 26.1  | 25.3  | 20.9  | 13.8  | 7.1   | 0.9  |
|                                                                                  | DBStd       | 10.74  | 6.29  | 6.41  | 6.36  | 5.14  | 4.46  | 3.39  | 3.12  | 3.27  | 4.17  | 5.16  | 5.64  | 6.01 |
|                                                                                  | HDD10.0     | 1177   | 337   | 252   | 131   | 28    | 2     | 0     | 0     | 0     | 0     | 22    | 119   | 286  |
|                                                                                  | HDD18.3     | 2707   | 594   | 479   | 349   | 170   | 55    | 4     | 0     | 1     | 21    | 157   | 337   | 541  |
|                                                                                  | CDD10.0     | 2319   | 2     | 7     | 42    | 118   | 264   | 408   | 501   | 476   | 326   | 140   | 32    | 4    |
|                                                                                  | CDD18.3     | 808    | 0     | 0     | 2     | 11    | 59    | 162   | 242   | 218   | 97    | 16    | 1     | 0    |
|                                                                                  | CDH23.3     | 8012   | 0     | 1     | 16    | 101   | 501   | 1552  | 2655  | 2197  | 858   | 129   | 3     | 0    |
| CDH26.7                                                                          | 3421        | 0      | 0     | 1     | 16    | 137   | 612   | 1274  | 1034  | 318   | 30    | 0     | 0     |      |
| Wind                                                                             | WSAvg       | 4.4    | 4.7   | 4.8   | 5.1   | 5.2   | 4.4   | 4.2   | 3.6   | 3.5   | 3.8   | 4.2   | 4.6   | 4.6  |
| Precipitation                                                                    | PrecAvg     | 1010   | 27    | 41    | 60    | 104   | 142   | 140   | 104   | 111   | 105   | 77    | 60    | 40   |
|                                                                                  | PrecMax     | 1433   | 79    | 99    | 151   | 229   | 309   | 241   | 372   | 255   | 284   | 188   | 188   | 111  |
|                                                                                  | PrecMin     | 618    | 2     | 0     | 22    | 17    | 35    | 33    | 20    | 25    | 12    | 4     | 6     | 1    |
|                                                                                  | PrecStd     | 182    | 18    | 26    | 33    | 46    | 72    | 63    | 74    | 60    | 60    | 48    | 45    | 26   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 17.6  | 21.2  | 25.5  | 28.8  | 32.2  | 35.9  | 38.5  | 38.7  | 35.9  | 29.7  | 23.6  | 18.7 |
|                                                                                  |             | MCWB   | 10.6  | 12.7  | 16.6  | 18.8  | 22.5  | 24.3  | 23.3  | 23.1  | 23.2  | 20.9  | 16.3  | 14.5 |
|                                                                                  | 2%          | DB     | 14.2  | 17.4  | 22.5  | 26.2  | 29.8  | 33.4  | 36.3  | 36.2  | 32.3  | 26.9  | 21.1  | 15.7 |
|                                                                                  |             | MCWB   | 9.2   | 11.9  | 15.2  | 17.6  | 21.2  | 23.9  | 24.5  | 24.2  | 22.9  | 18.8  | 15.1  | 12.0 |
|                                                                                  | 5%          | DB     | 11.3  | 14.0  | 20.0  | 24.0  | 28.0  | 32.1  | 34.5  | 33.9  | 30.3  | 24.1  | 18.1  | 12.4 |
|                                                                                  |             | MCWB   | 6.8   | 9.2   | 13.8  | 16.6  | 20.6  | 23.7  | 24.6  | 23.9  | 22.6  | 17.6  | 13.1  | 8.8  |
|                                                                                  | 10%         | DB     | 8.0   | 11.1  | 17.5  | 22.0  | 26.3  | 30.3  | 32.7  | 32.2  | 28.0  | 22.2  | 15.9  | 9.8  |
|                                                                                  |             | MCWB   | 4.5   | 7.0   | 12.6  | 15.9  | 19.6  | 22.9  | 24.4  | 23.7  | 21.1  | 17.1  | 11.8  | 6.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 13.6  | 15.2  | 18.6  | 20.5  | 24.1  | 26.4  | 27.5  | 26.8  | 25.2  | 22.8  | 18.3  | 15.8 |
|                                                                                  |             | MCDB   | 15.6  | 18.4  | 23.7  | 26.8  | 29.6  | 32.4  | 33.4  | 33.0  | 30.9  | 27.8  | 20.9  | 17.8 |
|                                                                                  | 2%          | WB     | 9.6   | 12.5  | 16.8  | 19.3  | 22.8  | 25.3  | 26.4  | 25.8  | 24.1  | 21.2  | 15.8  | 12.5 |
|                                                                                  |             | MCDB   | 13.2  | 16.3  | 20.7  | 24.1  | 28.0  | 31.7  | 33.0  | 32.4  | 29.8  | 24.7  | 19.2  | 14.6 |
|                                                                                  | 5%          | WB     | 7.1   | 9.9   | 15.1  | 17.9  | 21.7  | 24.4  | 25.7  | 25.0  | 23.3  | 19.5  | 14.3  | 9.2  |
|                                                                                  |             | MCDB   | 10.8  | 14.2  | 19.0  | 22.3  | 26.5  | 30.2  | 32.0  | 31.5  | 28.6  | 22.6  | 17.1  | 11.8 |
|                                                                                  | 10%         | WB     | 4.8   | 6.9   | 12.7  | 16.6  | 20.5  | 23.6  | 25.0  | 24.2  | 22.3  | 17.7  | 12.3  | 6.7  |
|                                                                                  |             | MCDB   | 8.3   | 10.6  | 16.7  | 21.3  | 25.1  | 28.9  | 31.1  | 30.2  | 27.0  | 20.5  | 15.5  | 9.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.2  | 10.5  | 11.3  | 11.6  | 10.9  | 10.6  | 10.9  | 11.3  | 11.5  | 11.3  | 10.8  | 9.8  |
|                                                                                  |             | MCDBR  | 14.7  | 15.5  | 14.8  | 14.1  | 12.5  | 12.0  | 13.0  | 13.7  | 12.9  | 13.3  | 13.7  | 13.7 |
|                                                                                  |             | MCWBR  | 10.1  | 10.5  | 8.8   | 8.1   | 6.2   | 5.0   | 4.4   | 4.7   | 5.6   | 7.0   | 8.8   | 9.9  |
|                                                                                  | 5% WB       | MCDBR  | 13.5  | 14.1  | 11.8  | 11.8  | 11.2  | 10.9  | 11.1  | 11.7  | 11.2  | 10.6  | 11.8  | 12.2 |
|                                                                                  |             | MCWBR  | 10.0  | 10.4  | 7.9   | 7.9   | 6.3   | 5.5   | 4.9   | 5.1   | 5.4   | 6.9   | 8.7   | 9.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.290  | 0.302 | 0.332 | 0.381 | 0.421 | 0.426 | 0.433 | 0.433 | 0.384 | 0.340 | 0.303 | 0.286 |      |
|                                                                                  | taud        | 2.479  | 2.453 | 2.426 | 2.308 | 2.251 | 2.268 | 2.275 | 2.277 | 2.366 | 2.477 | 2.523 | 2.515 |      |
|                                                                                  | Ebn at Noon | 894    | 928   | 928   | 896   | 862   | 855   | 845   | 836   | 860   | 870   | 871   | 873   |      |
|                                                                                  | Edh at Noon | 78     | 92    | 105   | 126   | 137   | 134   | 132   | 129   | 111   | 90    | 75    | 71    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.25   | 3.01  | 3.91  | 4.98  | 5.71  | 6.49  | 6.45  | 5.73  | 4.80  | 3.45  | 2.44  | 1.94  |      |
|                                                                                  | RadStd      | 0.23   | 0.34  | 0.33  | 0.32  | 0.51  | 0.47  | 0.41  | 0.31  | 0.33  | 0.39  | 0.24  | 0.18  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | -1.14   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (24 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page